Wai Yu Amanda Ng

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Highlight of Skills

- 3+ years research and industry experiences in data analysis to extract insights and support data-driven decision-making across multidisciplinary projects.
- Proficient in communicating analytical findings through data visualizations and stakeholder oriented reports.
- Experiences working in collaborative teams, strong problem-solving and critical thinking skills, attentive to details.
- Tools: SQL, R, Python, Excel, Tableau, Power BI, Git, QGIS
- Languages: English, Cantonese, Mandarin

Education

Sep 2025 – Apr 2026 (expected) **MSc in Statistics**, *University of Toronto*, Toronto, ON, Canada
Courses: Applied Multivariate Analysis, Methods of Applied Statistics, Special Topics in Statistical Data Science, Special Topics in Applied Statistics, Survival Analysis

Sep 2021 – Apr 2025 **H.B.Sc in Statistical Science (Specialist), Mathematics (Major), Health Studies (Focus)**, *University of Toronto*, Toronto, ON, Canada **cGPA: 3.94/4.00**
Courses: Applied Statistical Methods, Applied Spatial Analysis, Time Series Analysis, Surveys, Sampling and Observational Data, Statistical Consultation
Awards: Samuel Beatty In-course Scholarship (2025), University of Toronto Excellence Award (2024), Eleventh Annual Canadian Statistics Student Conference Undergraduate Level First Prize (2023), University of Toronto Dean's List Scholar (2021–2025)

Data-related Experiences

Sep 2025 – Current **R-Based Learning Designer**, *University of Toronto Human Biology Program*
- Developed 4 auto-testable R coding assignments for a cohort of 400 students, improving pedagogical effectiveness and reduced instructor grading workload.
- Designed 2 interactive LearnR modules with guided coding exercises and visualizations using RShiny, improving skills in reproducible coding, user-centered design, and data visualization.
- Collaborated with peer module developers to address issues and revise updated modules, strengthening teamwork and technical communication skills.

Sep 2025 – Dec 2025 **Statistical Neuroimaging Researcher**, *University of Toronto, Department of Statistical Sciences*
- Investigated statistical inference for genetic heritability of intermodal coupling using simulated multi-modal brain MRI data, providing insights for advanced methodological approaches in neuroimaging data.
- Conducted 50+ large-scale simulation experiments and applied adaptive clustering techniques in R with parallel computing, validating proposed model's ability in controlling Type I error rates and statistical power.
- Summarized findings in a detailed report, providing evidence that informed refinement of current methods and shaped the development of more robust neuroimaging statistical approaches.

May 2024 – Current **Regression Analysis Researcher**, *University of Toronto Department of Statistical Sciences*
- Conducted research on robust regression methods for handling outliers and heavy-tailed data, resulting in a preprint paper and a departmental research day presentation.
- Developed and evaluated 4 parameter estimation approaches under high-dimensional and outlier settings, generating a comprehensive performance analysis that provided insights for improving traditional estimation accuracy.
- Implemented and troubleshot optimization algorithms to address non-convergence issues and enhanced computational efficiency, enhancing proficiency for statistical computing through utilizing software R and Python.
- Presented research findings through detailed reports and visualizations and published an R package with 20+ reusable functions, strengthening statistical communication and facilitate research reproducibility.

Sep 2024 – Apr 2025 **Machine Learning Researcher**, *University of Toronto Department of Statistical Sciences*

- Investigated fairness in machine learning with biased labels in healthcare data extracted from MIMIC-III clinical database consisting of 40,000+ ICU patient records, leading to a departmental research day presentation.
- Evaluated embeddings-building algorithms performance in capturing semantic relationships among 6,000+ medical concepts, uncovering key predictors that enhanced downstream modeling.
- Trained supervised regression and random forest models for depression diagnosis and conducted fairness evaluations across demographic groups, revealing disparities in prediction outcomes linked to biased label distributions.
- Maintained a reproducible workflow in R and curated a GitHub repository to share code and results, strengthening transparency and enabling replication by other researchers.

Sep 2024 – Apr 2025 **Data Visualization Analyst**, *University of Toronto Human Biology Program*

- Analyzed student learning resources engagement data from Quercus and course grades using Excel, results informed data-driven evidence-based teaching strategies for course improvement.
- Created 100+ Tableau visualizations that transformed complex datasets into clear, compelling insights for faculty, showcasing advanced data visualization proficiency and strong statistical communication skills.
- Collaborated with peers to troubleshoot data challenges and refined visualization approaches, strengthening problem-solving and teamwork skills.

Jun 2024 – Apr 2025 **Psychiatric Education Quantitative Researcher**, *Centre for Addiction and Mental Health*

- Conducted research on a study investigating relationship between physical movement and cybersickness symptoms in 2 immersive VR medical trainings, resulting in a peer-reviewed publication and 2 poster presentations.
- Summarized descriptive statistics and performed statistical analyses in R, uncovering a significant correlation between nausea severity and the extent of movement.
- Provided data-driven recommendations for the design of VR training programs that prioritize user comfort, emphasizing the importance of movement considerations while maintaining educational efficacy.

Sep 2023 – Apr 2024 **Agent-Based Modeling Researcher**, *University of Toronto COBWEB lab*

- Initiated research using agent-based simulation to investigate declining birth rates and potential solutions, resulting in a poster presentation.
- Conducted comprehensive literature reviews and prepared an annotated bibliography of 30+ sources, synthesizing multidisciplinary findings that improved study design and strengthened critical thinking and organizational skills.
- Mastered the simulation software COBWEB to model population dynamics and policy scenarios, independently developing 20+ simulations that expanded research scope and demonstrated advanced self-learning ability.
- Performed time-series and regression analyses in Python to assess the impact of policy interventions on birth rates trends, providing insights for short-term and long-term strategic proposal.

May 2023 – Aug 2023 **Data Analyst Intern**, *Canadian Urban Institute*

- Developed regression models in R to analyze nationwide household and business-level carbon emissions and social connection patterns, providing data-driven insights for an ongoing Canadian city planning sustainability project.
- Created an interactive tool to demonstrate the effects of urbanization and housing composition on greenhouse gas emissions, enabling stakeholders to model scenarios for policy planning.
- Utilized geospatial visualization tool QGIS to produce 300+ detailed maps, combined with statistical findings, revealing urban planning insights and actionable recommendations for decision-makers.

Oct 2022 – May 2023 **Computational Social Science Researcher**, *University of Toronto Population Well-being Lab*

- Applied machine learning methods (Gaussian Mixture Model, k-means, and hierarchical clustering) to survey data consisting of 150k+ observations in R, to uncover meaningful subpopulations based on life satisfaction and demographics, resulting in 3 conference presentations.
- Prepared detailed 4 tip-sheets on algorithms and evaluated effectiveness and appropriateness of statistical methods, enhancing team understanding in methodological application.
- Designed a conference poster discussing the suitability of clustering methods and implications of results, effectively explaining complex workflow for diverse audiences, resulting in a conference poster first prize.

Sep 2022 – Dec 2023 **Statistical Consultant**, *University of Toronto Data Sciences Café*

- Contributed to a supportive and collaborative consulting community, fostering an environment where interdisciplinary research and statistical analysis could thrive.
- Recommended tailored statistical methodologies and identified limitations in proposed study designs to 20+ clients, delivering actionable insights that improved methodological validity and analytical outcomes across diverse fields.
- Identified potential biases and limitations in proposed research methods, providing actionable insights for improvement in study method design.
- Prepared 5 reports on advanced statistical topics, introducing statistical concepts and coding methods in an accessible manner to enhance statistical literacy among non-statistician researchers.

Statistics Course Projects

Oct 2025 – Dec 2025 **Uncovering Calendar-Based Structures Through Clustering of Daily Traffic Volume Profiles**, *STA2005*

- Applied PCA and UMAP dimensionality reduction techniques in R to year-long traffic volume data across 26 locations, uncovering recurring temporal clusters tied to school calendars, holidays, and seasonal routines.
- Delivered actionable insights for transportation planning and infrastructure maintenance by visualizing and communicating traffic flow patterns to non-technical stakeholders.

Oct 2025 – Dec 2025 **Fatal US School Shootings (1999-2025)**, *STA4101*

- Modeled fatality risk in 400+ US school shootings using logistic regression, highlighting actionable risk indicators including shooting intent, absence of non-shooter injuries, and shooter connection to the school community.
- Developed a reproducible GitHub repository with version control documenting all data cleaning, modeling, and analysis steps for transparency and collaboration.
- Prepared a written report synthesizing statistical findings into clear, evidence-based insights, informing safety and emergency preparedness strategies.

Oct 2024 – Dec 2024 **Transit and Crime: Investigating Assault Clusters Around Toronto's TTC Subway Lines**, *STA465*

- Conducted spatial analysis of 200k+ Toronto assault incidents using kernel density estimation, point process models and HDBSCAN clustering, successfully identified assault concentration patterns within varying buffer distances around TTC subway infrastructure.
- Developed interactive geospatial report with maps translating statistical findings into actionable insights for urban planners and policymakers on transit-related public safety.

Sep 2024 – Apr 2025 **Quercus Engagement as a Predictor of First-Year University Student Dropout Risk: A Logistic Regression Analysis**, *STA490*

- Leveraged R to preprocess and analyze engagement data from 3,000+ first-year students, improving data quality by resolving missing values and standardizing predictors across multiple time windows.
- Built and validated logistic regression models to assess the relationship between Quercus engagement and dropout risk, identifying early disengagement in first 4 weeks as a critical retention indicator.
- Summarized findings in stakeholder-focused reports with clear visualizations and accessible interpretations, providing evidence that informed refinement of learning support strategies.

Health Studies Course Projects

Feb 2024 – Apr 2024 **Effectiveness of Usage of Virtual Reality in Raising Awareness about Cannabis Misuse Issues among Teenages**, *HMB342*

- Designed a randomized controlled trial study comparing virtual reality–based education with traditional seminars to evaluate effectiveness in raising awareness of cannabis misuse among Canadian high school students.
- Developed evaluation metrics and statistical analysis plan (pre/post surveys, T-tests, knowledge retention, behavioral intentions) to measure intervention impact on awareness and future cannabis use.
- Applied health promotion and program design frameworks to integrate emerging technologies into adolescent substance misuse education, highlighting cost-effectiveness and policy implications for public health strategies.

Feb 2023 – Apr 2023 **Food Insecurity Intervention Program Planning**, *HST330*

- Conducted root cause analysis and developed a program proposal to address food insecurity among Toronto low-income families, modeling intervention design on the Supplemental Nutrition Assistance Program with a digital e-coupon system.
- Designed a quasi-experimental evaluation plan, including baseline and post-intervention measures (Food Insecurity Experience Scale, BMI, Healthy Eating Index, stress levels) to assess program effectiveness.
- Created a logic model and theory of change framework to outline expected outcomes, health implications, and long-term policy applications, demonstrating strong skills in program evaluation and health policy analysis.

Oct 2022 – Dec 2022 **Proposal to Address Teenager Substance Misuse Problem**, *HST209*

- Conducted comprehensive analysis of risk factors (family influence, peer pressure, media exposure, mental health) contributing to teenage substance misuse, synthesizing evidence-based research into actionable insights.
- Designed and evaluated multi-level interventions including individual counseling, family therapy, in-school workshops, and peer-led support communities to promote adolescent health and reduce substance misuse.
- Developed health promotion strategies grounded in empowerment and community engagement, aligning interventions with public health models to improve long-term outcomes for at-risk youth.

Other Experiences

Sep 2023 – Current **Teaching Assistant and Learning Support**, *University of Toronto Department of Statistical Sciences and Data Sciences Institute*

DoSS Courses: STA130, STA238, STA304, Statistics Aid Centre, Florence Nightingale Day (2024, 2025)
DSI Modules: SQL, Applying Statistical Concepts, Sampling, Visualization

- Prepared tutorials and office hours for statistics courses, strengthening programming (R, Python, SQL) skills and addressed questions across a wide range of statistical courses, enhancing students' statistical learning.
- Engaged with high school students at events such as Florence Nightingale Day, translating high-level statistical concepts into accessible and engaging explanations to spark interest in statistics.

Sep 2022 – Apr 2023 **First-year Learning Community Assistant Peer Mentor**, *University of Toronto*

- Facilitated 12 sessions to guide 20+ first year students in academic matters, fostering their skills to overcome challenges and manage transition anxiety to university life.
- Established a rapport and inclusive community among students, promoting a supportive learning environment.
- Coordinated discussions with Mathematics professors and library staff and managed online Quercus page to facilitate remote communication and engagement with students.

Volunteer

Sep 2022 – Dec 2022 **Data Entry Volunteer**, *St. Michael's Hospital Trauma Research Program and Injury Prevention*

- Reviewed eligible medical charts to collect injury/poisoning information
- Documented medical records into the system